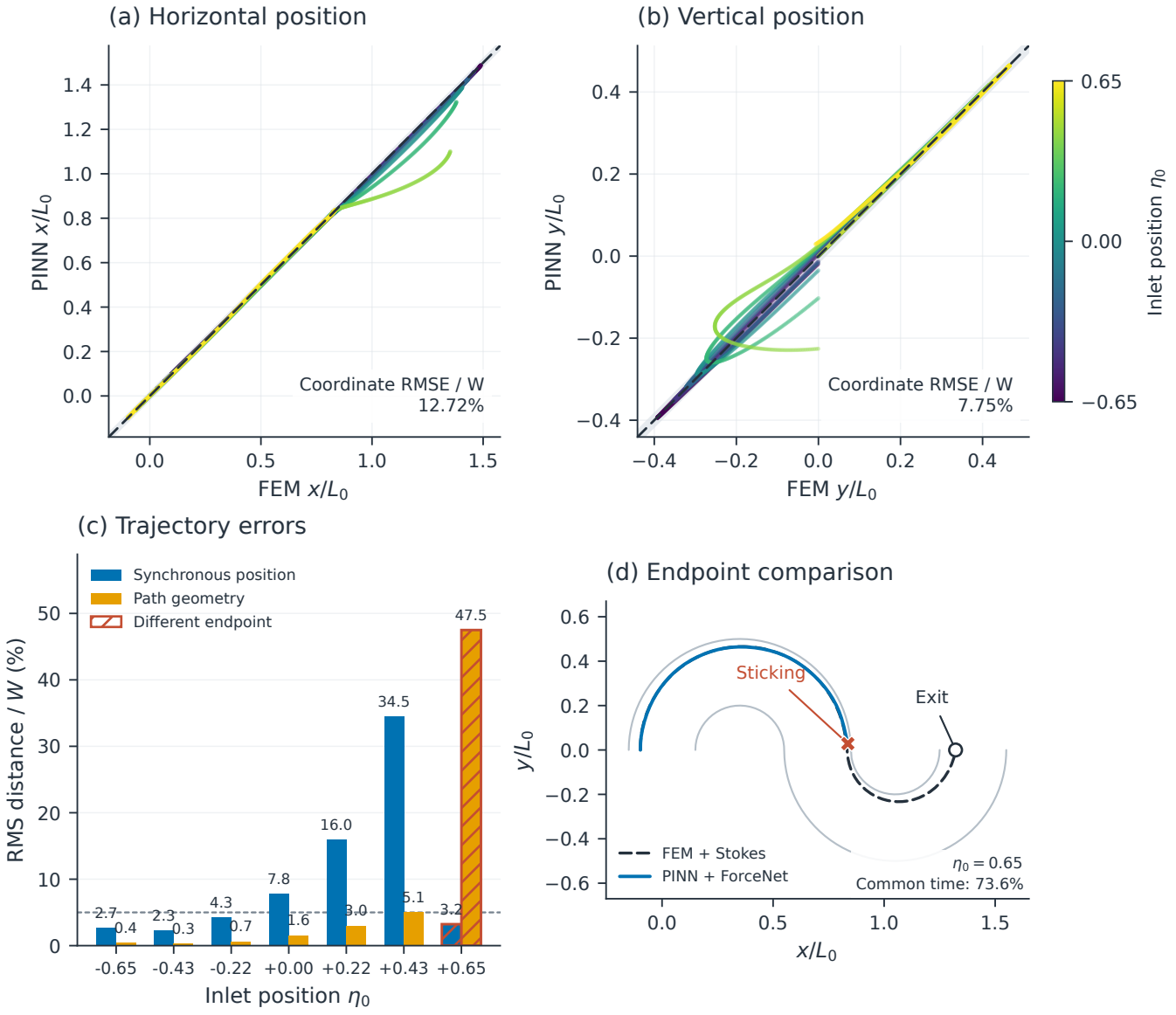
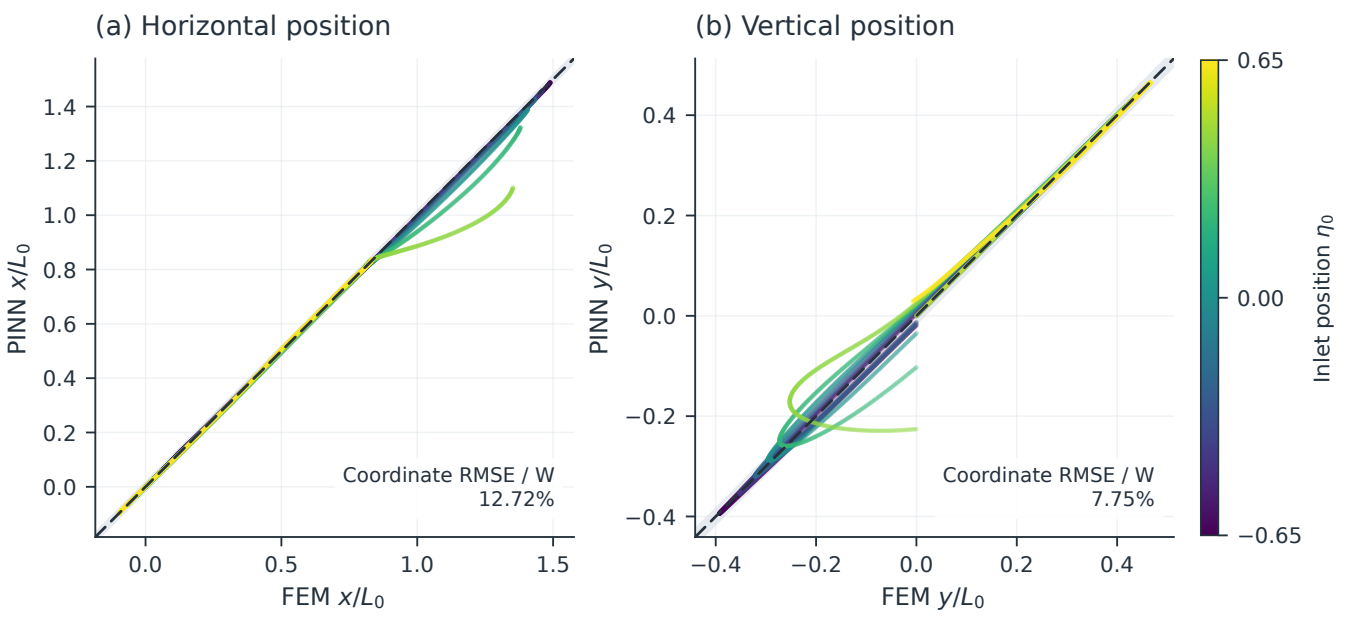
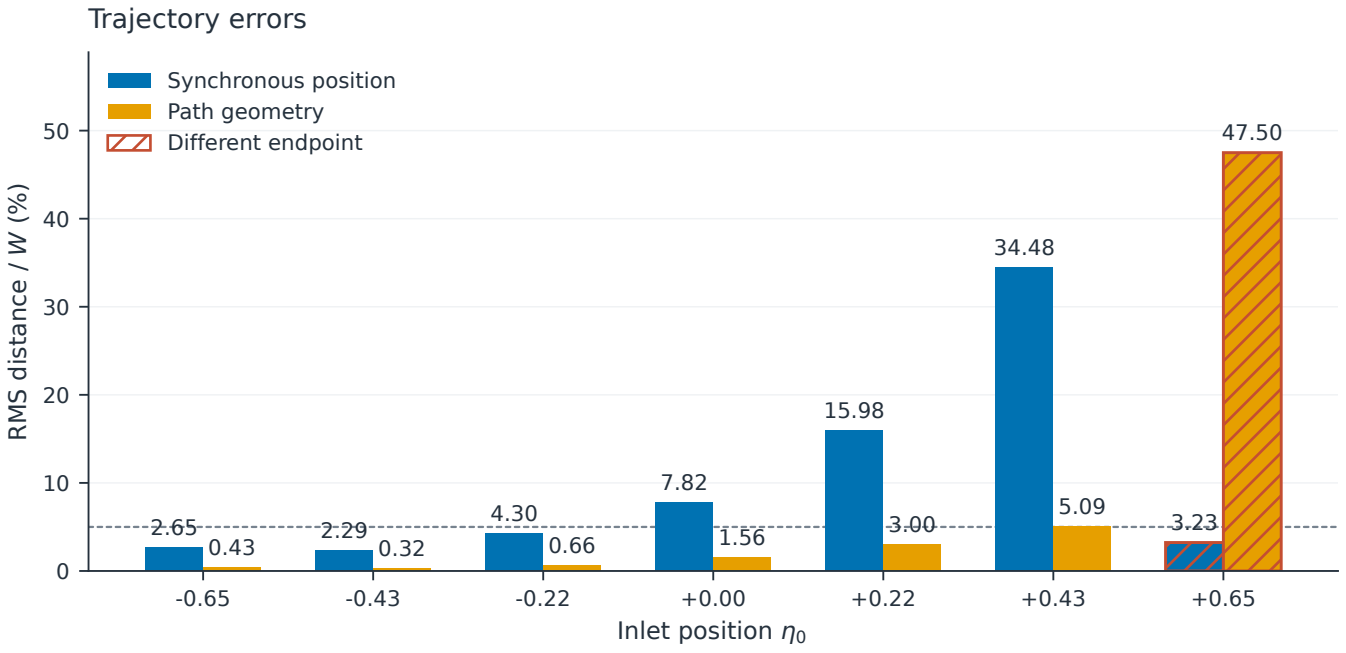
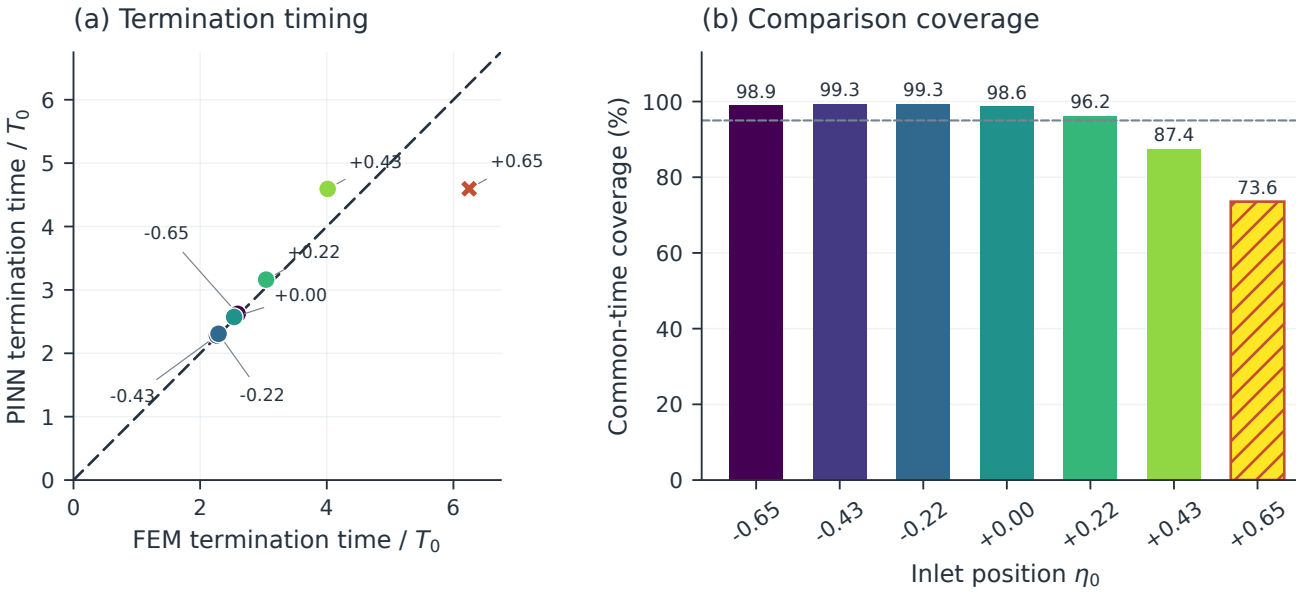
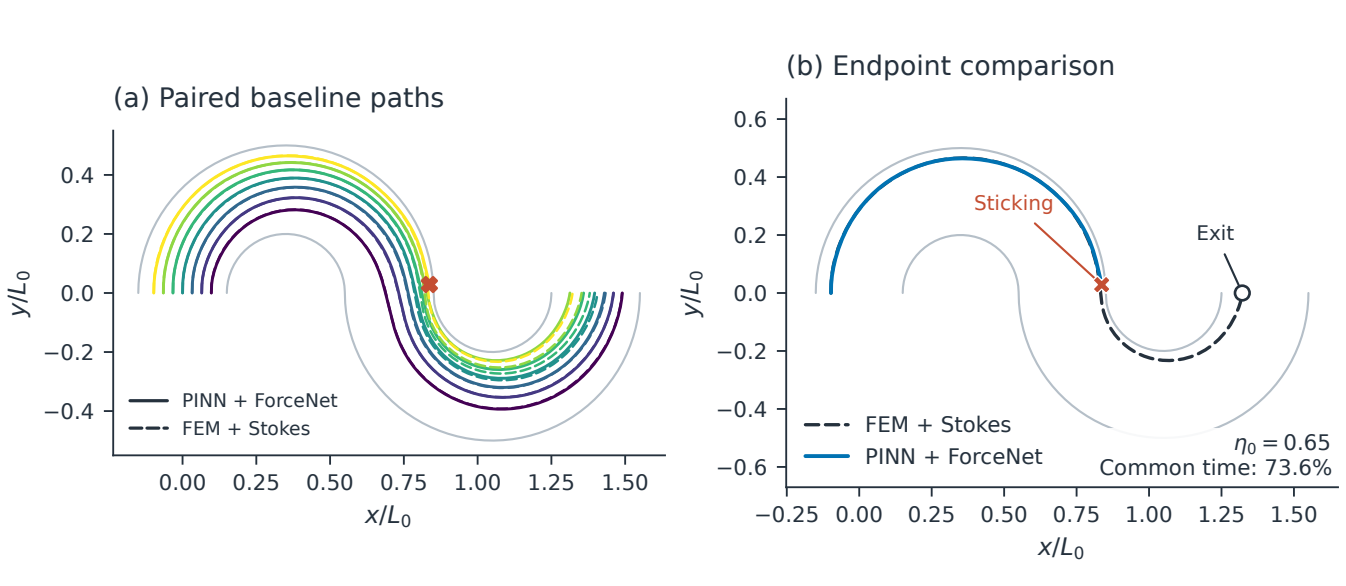
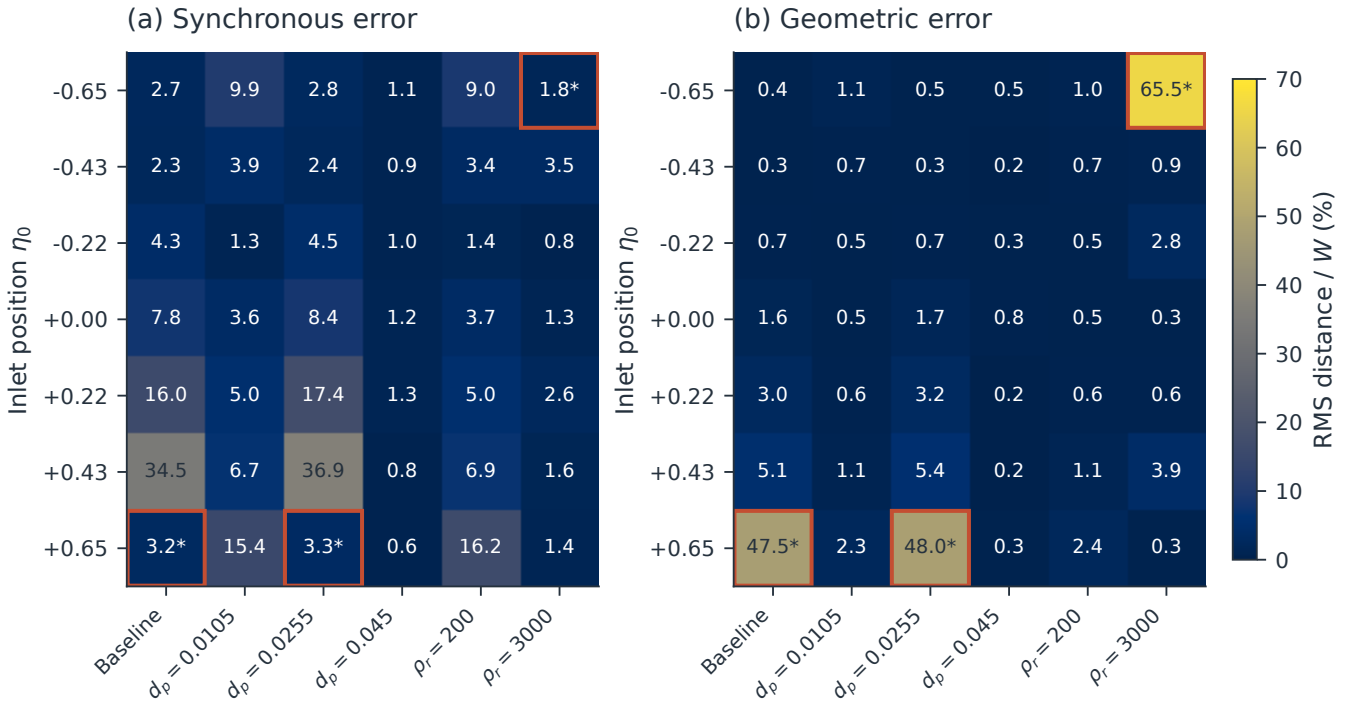

(a) Horizontal position
PINN x/L_0
FEM x/L_0
Coordinate RMSE / W
12.72%
(b) Vertical position
PINN y/L_0
FEM y/L_0
Coordinate RMSE / W
7.75%
Inlet position η_0
0.65
0.00
−0.65
(c) Trajectory errors
Synchronous position
Path geometry
Different endpoint
RMS distance / W (%)
Inlet position η_0
-0.65
-0.43
-0.22
+0.00
+0.22
+0.43
+0.65
2.7
0.4
2.3
0.3
4.3
0.7
7.8
1.6
16.0
3.0
34.5
5.1
3.2
47.5
(d) Endpoint comparison
Sticking
Exit
FEM + Stokes
PINN + ForceNet
$\eta_0 = 0.65$
Common time: 73.6%
y/L_0
x/L_0

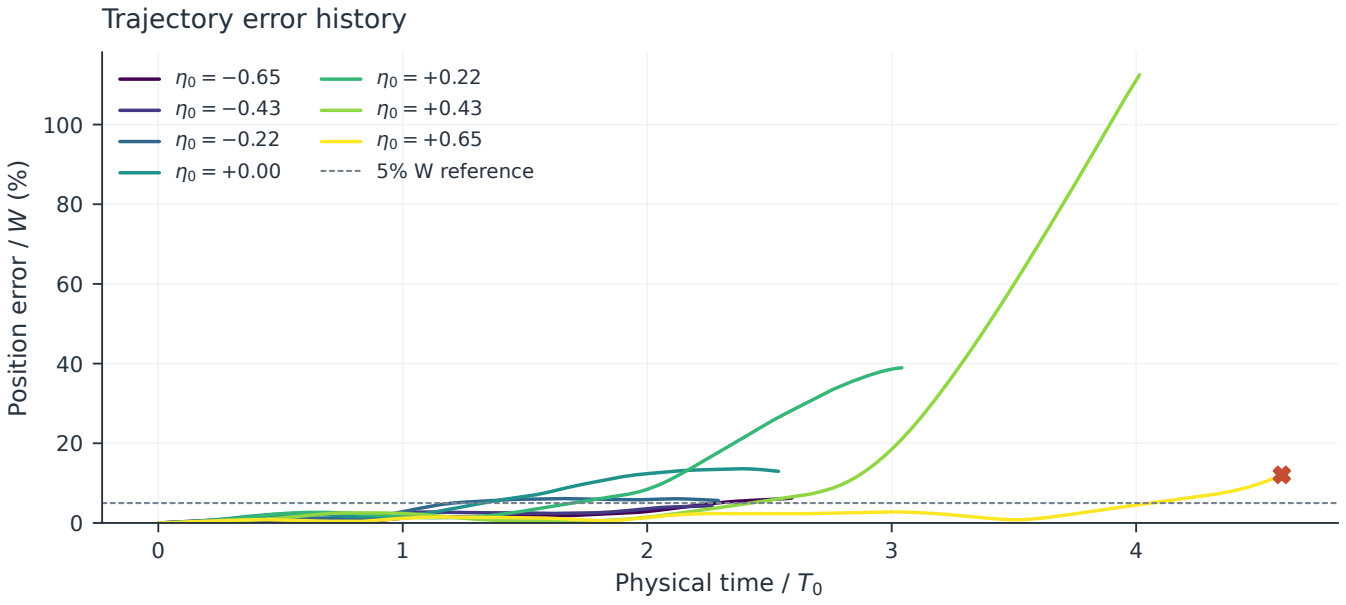

(a) Horizontal position
PINN x/L_0
FEM x/L_0
Coordinate RMSE / W
12.72%
(b) Vertical position
PINN y/L_0
FEM y/L_0
Coordinate RMSE / W
7.75%
Inlet position η_0
0.65
0.00
−0.65


Trajectory errors
Synchronous position
Path geometry
Different endpoint
RMS distance / W (%)
0
10
20
30
40
50
2.65
0.43
2.29
0.32
4.30
0.66
7.82
1.56
15.98
3.00
34.48
5.09
3.23
47.50
-0.65
-0.43
-0.22
+0.00
+0.22
+0.43
+0.65
Inlet position η_0


(a) Termination timing
PINN termination time / T_0
FEM termination time / T_0
0
1
2
3
4
5
6
+0.43
+0.65
-0.65
+0.22
+0.00
-0.43
-0.22
(b) Comparison coverage
Common-time coverage (%)
0
20
40
60
80
100
98.9
99.3
99.3
98.6
96.2
87.4
73.6
-0.65
-0.43
-0.22
+0.00
+0.22
+0.43
+0.65
Inlet position η_0


(a) Paired baseline paths
0.4
0.2
0.0
−0.2
−0.4
y/L_0
0.00
0.25
0.50
0.75
1.00
1.25
1.50
x/L_0
PINN + ForceNet
FEM + Stokes
(b) Endpoint comparison
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
y/L_0
Sticking
Exit
FEM + Stokes
PINN + ForceNet
$\eta_0 = 0.65$
Common time: 73.6%
−0.25
0.00
0.25
0.50
0.75
1.00
1.25
1.50
x/L_0


(a) Synchronous error
(b) Geometric error
Inlet position η_0
-0.65
-0.43
-0.22
+0.00
+0.22
+0.43
+0.65
2.7 9.9 2.8 1.1 9.0 1.8*
2.3 3.9 2.4 0.9 3.4 3.5
4.3 1.3 4.5 1.0 1.4 0.8
7.8 3.6 8.4 1.2 3.7 1.3
16.0 5.0 17.4 1.3 5.0 2.6
34.5 6.7 36.9 0.8 6.9 1.6
3.2* 15.4 3.3* 0.6 16.2 1.4
0.4 1.1 0.5 0.5 1.0 65.5*
0.3 0.7 0.3 0.2 0.7 0.9
0.7 0.5 0.7 0.3 0.5 2.8
1.6 0.5 1.7 0.8 0.5 0.3
3.0 0.6 3.2 0.2 0.6 0.6
5.1 1.1 5.4 0.2 1.1 3.9
47.5* 2.3 48.0* 0.3 2.4 0.3
Baseline
$d_\rho = 0.0105$
$d_\rho = 0.0255$
$d_\rho = 0.045$
$\rho_r = 200$
$\rho_r = 3000$
RMS distance / W (%)
0
10
20
30
40
50
60
70
* Different endpoint class; repeated baseline control omitted


Trajectory error history
$\eta_0 = -0.65$
$\eta_0 = -0.43$
$\eta_0 = -0.22$
$\eta_0 = +0.00$
$\eta_0 = +0.22$
$\eta_0 = +0.43$
$\eta_0 = +0.65$
5% W reference
Position error / W (%)
Physical time / T_0
0
20
40
60
80
100
0
1
2
3
4